Comparison of Model Performance on Classifying Driving Behaviors

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Abstract

Driver behavior classification plays an increasingly important role in automotive safety. Using accelerometer and gyroscope data, this project addresses a supervised multi-class classification task aimed at detecting three behavior categories, SLOW, NORMAL, and AGGRESSIVE. This work evaluates four traditional machine learning methods (Logistic Regression, Random Forest, XGBoost, and LightGBM) using a feature-engineering pipeline with windowed temporal segmentation, statistical descriptors, and derivative-based features. Performance is assessed using multiple metrics including accuracy, weighted precision, recall, F1-score, and loss to capture both classification quality and confidence. The final comparison demonstrates XGBoost achieves the highest weighted F1 score on the test set, while Random Forest and LightGBM performed competitively. Optimization efforts include smoothing raw noise, SMOTE balancing, hyperparameter tuning, and class-weight adjustments.

Introduction

Aggressive driving is closely associated with elevated accident severity and fatal crash likelihood. Accurate detection of driving style from sensor data, particularly accelerometer and gyroscope readings, provides opportunities for safety interventions, driver training, and risk-based insurance modeling.

The objective of this project is to build a machine learning system capable of classifying driving segments into three categories:

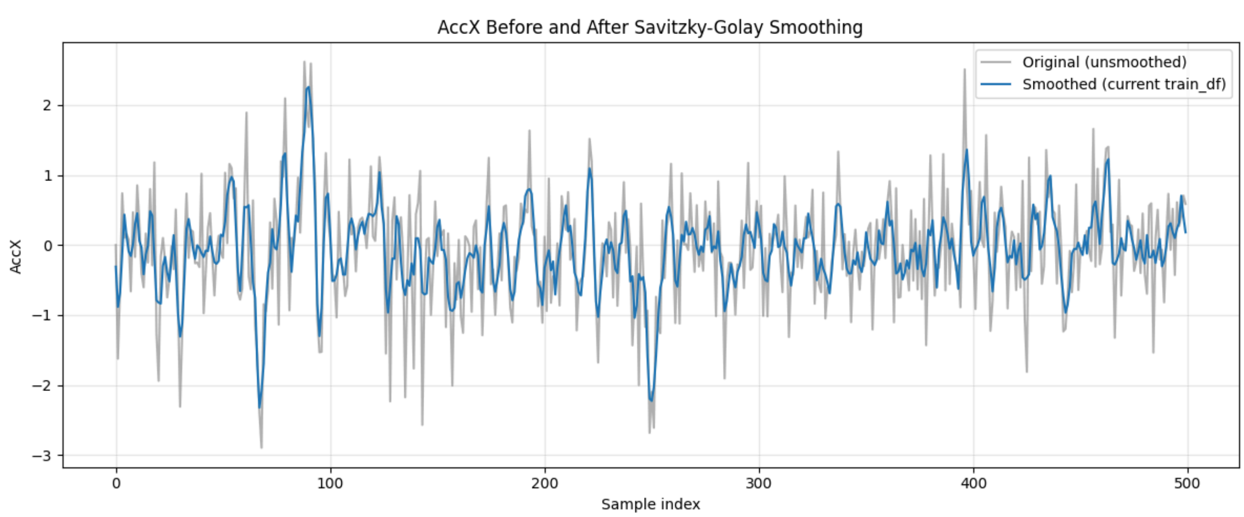
- SLOW
- NORMAL
- AGGRESSIVE

The dataset consists of motion sensor data collected from a smartphone, with six continuous features (AccX, AccY, AccZ, GyroX, GyroY, GyroZ) and a timestamp. The task involves challenges typical of mobile sensor classification, including sensor noice, class imbalance, and temporal dynamics. The goal is to design a high-performance pipeline consistent with project requirements while presenting high scores.

Methods

Data Preparation:

The dataset was loaded and explored, revealing that training samples had 3644 rows, and testing samples had 3084 rows. Raw signals exhibit noise, spikes, and jitter. To compensate for this, a Savitsky-Golay filter was applied independently to each sensor axis:



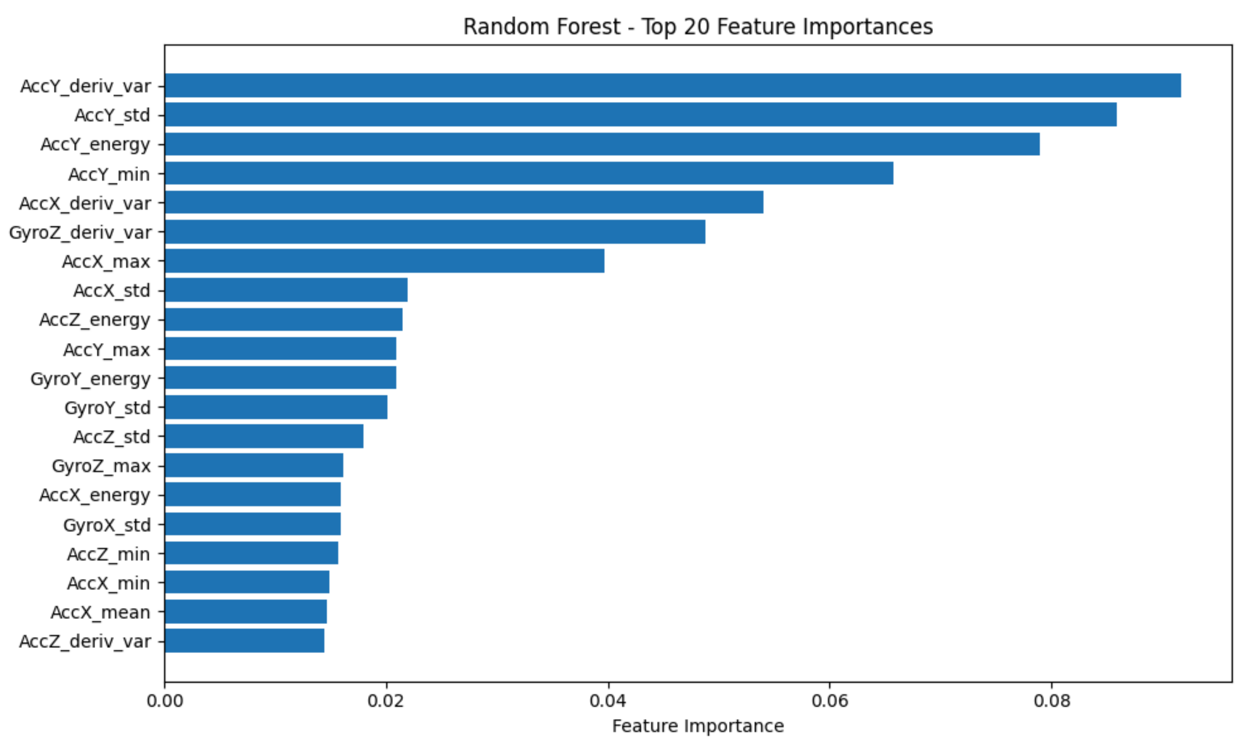
This smoothing preserves major trends while reducing random signal fluctuations.

Windowing for Temporal Context:

Because single timestamp values are not meaningful indicators of driving style, data was segmented into overlapping windows, with a size of 100 samples and a stride of 20. This produced sequences capturing short temporal intervals.

Feature Engineering:

To feed traditional ML models, each window was converted into a feature vector using statistical descriptors. For each axis, mean, standard deviation, min/max range, skewness, and kurtosis. This transformation yielded interpretable features summarizing movement patterns.



Resampling and Scaling:

The dataset exhibits class imbalance, with the NORMAL class often dominating. To address this, SMOTE oversampling was applied to feature-engineered training data and StandardScaler was fitted to training windows and testing data.

Model Training:

Four models were trained on the feature-engineered and resampled training set:

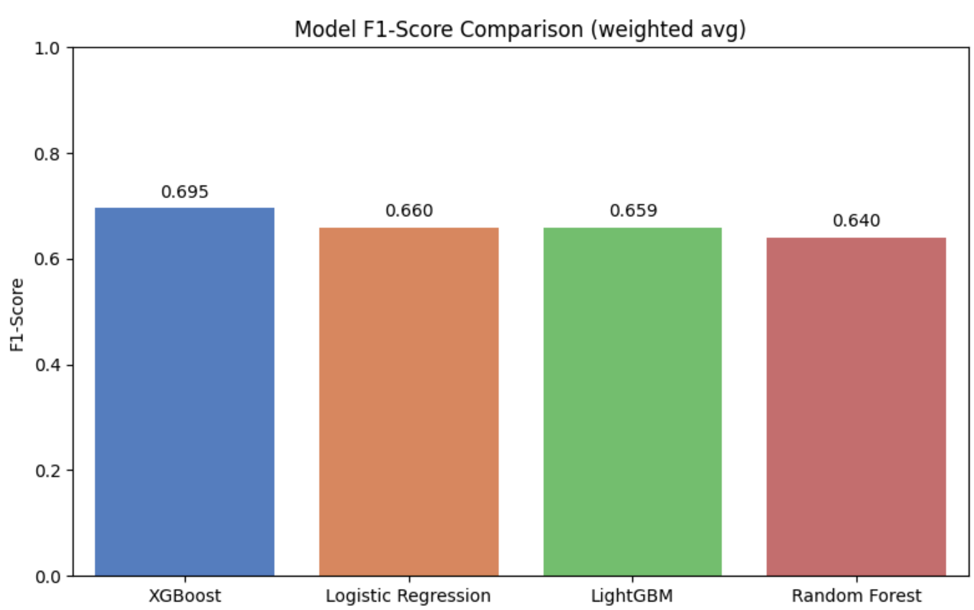
- Logistic Regression:
  - class_weight='balanced'
  - max_iter=1000
- Random Forest:
  - n_estimators=500
  - max_depth=40
  - min_samples_split=5
  - min_samples_leaf=2
  - max_features='sqrt'
  - class_weight='balanced_subsample'
  - random_state=42
- XGBoost
  - n_estimators=600
  - max_depth=6
  - learning_rate=0.1
  - subsample=0.9

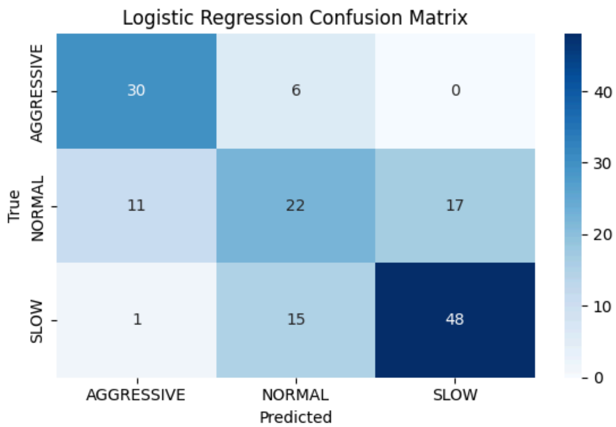
- colsample_bytree=0.8
- gamma=0.1
- eval_metric='mlogloss'

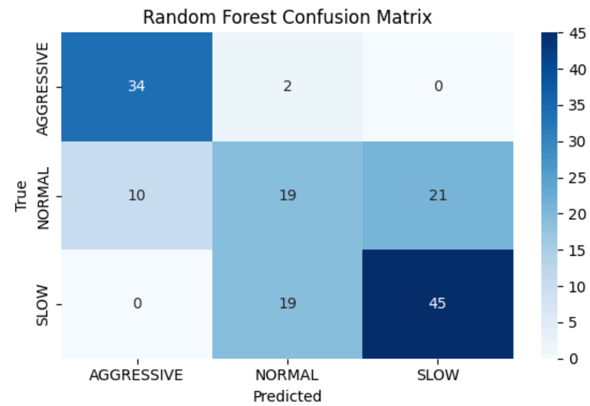
- LightGBM
  - learning_rate=0.05
  - num_leaves=63
  - feature_fraction=0.8
  - bagging_fraction=0.8
  - min_child_samples=10
  - objective='multiclass'
  - metric='multi_logloss'
  - is_unbalance=True

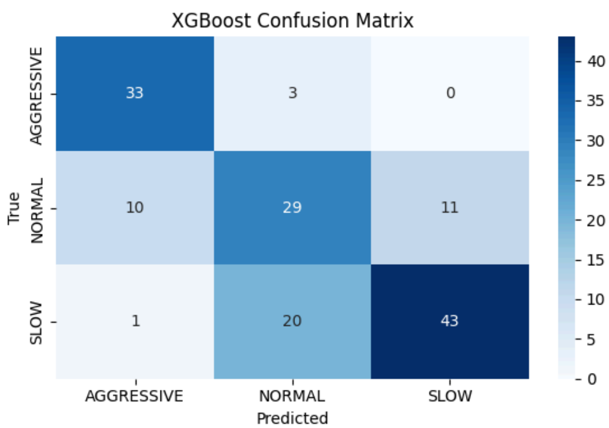
Evaluation Metrics:

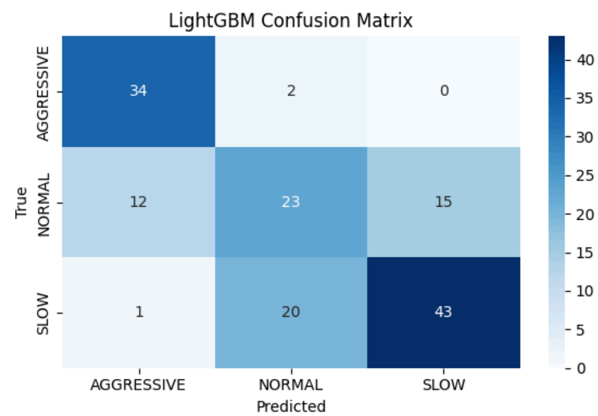
Models were evaluated on the test set using accuracy, weighted precision, weighted recall, weighted F1-score, log loss, and confusion matrices.











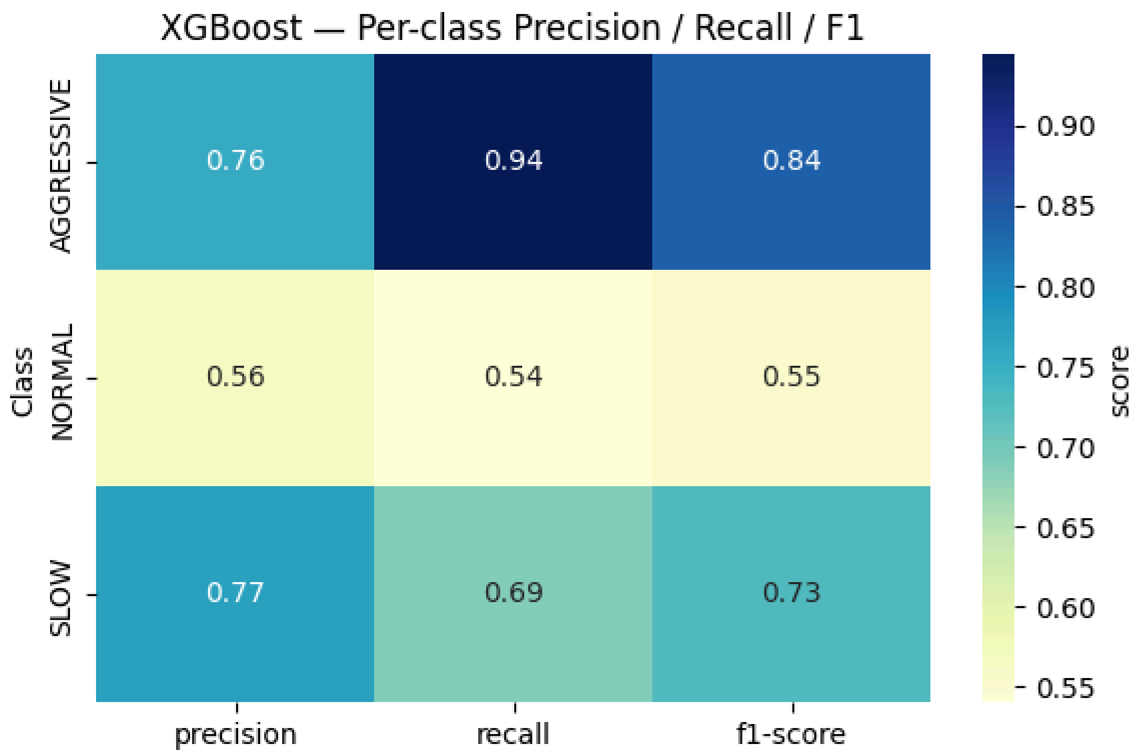
Results

Below is a summary of the final computed metrics:

Model	Precision	Recall	F1-Score	Accuracy	Log Loss
Logistic Regression	0.657048	0.666667	0.659841	0.666667	0.987407
Random Forest	0.634697	0.653333	0.640125	0.653333	0.662583
XGBoost	0.698190	0.700000	0.695458	0.700000	0.800591
LightGBM	0.660309	0.666667	0.658795	0.666667	0.841273

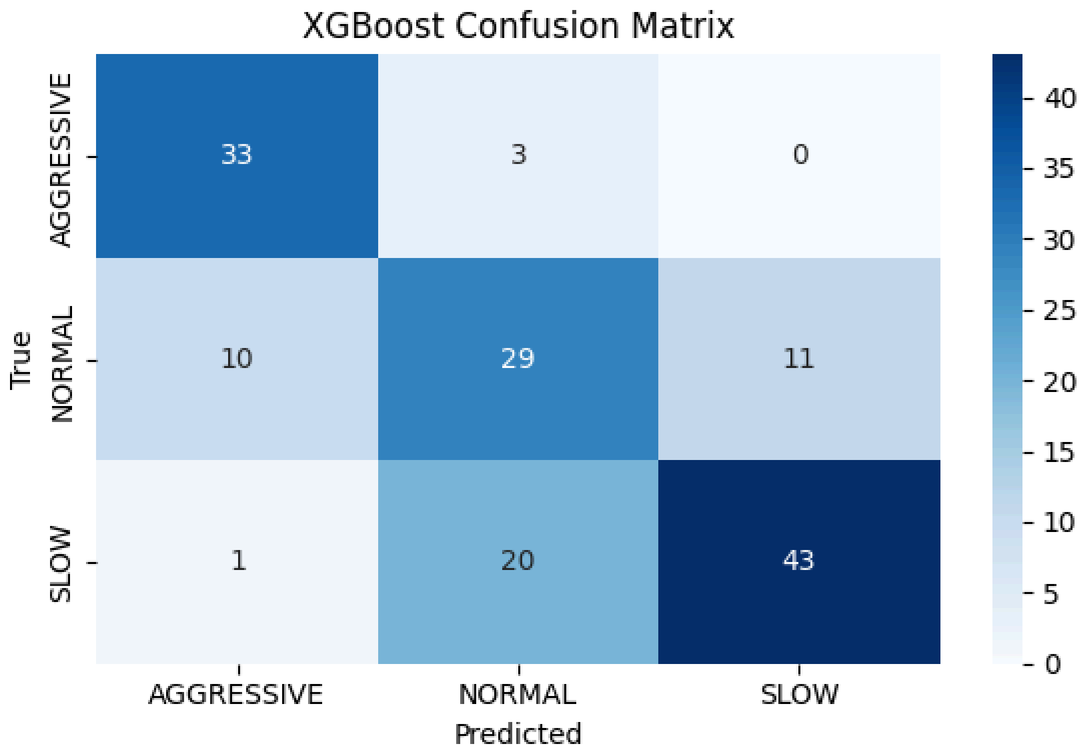
Model Comparison:

XGBoost achieved the highest weighted F1 score and accuracy. Logistic Regression and LightGBM performed similarly but lower, and Random Forest had the best log loss.



Confusion Matrix Insights:

Observations from confusion matrices include that SLOW and AGGRESSIVE classes were predicted more consistently. NORMAL was the most difficult for all models because it was confused with both extremes. This class imbalance contributed to misclassfication.



Discussion

Successes:

- Enhanced feature engineering significantly improved detection of motion patterns.
- SMOTE resampling reduced model bias toward the majority class.
- Hyperparameter tuning yielded meaningful improvements, particularly for XGBoost and Random Forest.

Limitations:

- Window-based segmentation may lose long-term temporal context.
- The NORMAL class remains challenging, possibly due to overlapping motion characteristics.
- Sensor noise and variability in driving conditions limits the ceiling of classification accuracy.

The comparison of multiple models revealed that tree-based ensembles methods, particularly XGBoost, outperformed linear models such as Logistic Regression. Other ensemble methods like Random Forest and LightGBM also performed competitively. This outcome aligns with expectations given the non-linear nature of motion sensor data. Driving behavior inherently involves complex relationships between acceleration, rotation, and patterns that ensemble models can capture more effectively through non-linear decision boundaries.

Among all tuned models, XGBoost performed the best with the highest scores across all metrics, including a 70% accuracy score. Adding a balanced class weight parameter helped the models handle underrepresented classes, smoothing out a bias toward the majority driving style. Also, the XGBoost's gradient boosting framework incrementally corrected previous classification errors, allowing it to handle small variations in motion features. With a larger dataset size and more tuning, I believe that the scores for XGBoost and other ensemble models could rise.

Optimization Recommendations

Further improvements are possible, including:

- Adjusting ranges and refining hyperparameters like gamma, subsample for XGBoost, and min_samples_leaf, max_depth for Random Forest.
- Feature selection/pruning, removing weakly contributing features.
- Window optimization, tuning window size and stride.
- Additional derived features like jerk, magnitude norms, etc.

Conclusion

This project demonstrates that traditional machine learning methods, combined with data processing and feature engineering, can effectively classify driving behavior from smartphone data. Among all tested classifiers, XGBoost yielded the best overall performance, followed closely by Random Forest and LightGBM.

Despite challenges with class imbalance and overlapping behavioral classes, results show strong potential for real-world applications, such as driver monitoring and safety analytics. Future improvements may include further optimized windowing, testing more advanced temporal models or deep learning methods.